

Firmware, BIOS, UEFI and Drivers — Detailed Lecture Notes

JKSSB Computer Awareness

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46.1 What firmware is

Every computer has two broad parts: the hardware, which is the physical circuitry, and the software, which is the set of instructions. Between them sits a special kind of software called **firmware**. Firmware is software that is stored in non-volatile memory on the hardware itself and that gives the device its basic, low-level instructions. It is not opened by the user like a word processor or a web browser; it is fixed in the device and runs by itself.

Firmware runs *before* the operating system. When a computer is switched on, its memory is empty and no program is running, so something must tell the hardware how to begin. That “something” is the firmware. It starts the hardware, checks it, and then hands control to the operating system.

Definition (Firmware). *Software stored in non-volatile memory that is programmed into a hardware device and provides its basic, low-level startup and control instructions, running before the operating system.*

Firmware is different from ordinary application software in three ways. It is stored on the device rather than installed like a program; it is tied closely to the exact hardware; and it starts the machine rather than being started by the user.

46.2 Where firmware is stored

Firmware must be available the instant the machine is switched on, so it is kept in **non-volatile** memory — memory that keeps its contents without power. If firmware were stored in RAM, it would be lost the moment the power was switched off, and the machine could never start.

Common homes for firmware are **ROM** (Read-Only Memory), **PROM** (Programmable ROM), **EPROM** (Erasable PROM), **EEPROM** (Electrically Erasable PROM), and **flash memory**. Flash memory is the modern choice because it can be erased and rewritten while still installed, which is what makes firmware updates possible.

Definition (Non-volatile memory). *Memory that retains its contents when the power is removed. ROM, flash memory, an SSD, and an HDD are non-volatile. RAM is the common volatile memory.*

The important contrast is with RAM. RAM is volatile and is empty at power-on; firmware is non-volatile and is present at power-on. A common wrong statement is that firmware is loaded into RAM before the machine starts — in fact the firmware resides in ROM/flash and runs from there.

It is worth naming the members of the ROM family, because exam papers often ask which memory type can be rewritten. **ROM** is programmed at the factory and cannot be changed. **PROM** (Programmable ROM) can be written once by the user. **EPROM** (Erasable PROM) can

be erased with ultraviolet light and reprogrammed. **EEPROM** (Electrically Erasable PROM) can be erased and rewritten electrically. **Flash memory** is a fast, block-erasable form of EEPROM and is the type used for modern firmware, which is why firmware can be updated in place.

Example 46.2.1. *An item states, “The firmware of a modern PC is stored in RAM so that it can be updated.”: This is wrong on the first point. Firmware is stored in non-volatile ROM or flash memory, not in volatile RAM. Flash memory is what makes the firmware updatable, not RAM.*

46.3 Hardware, software, and firmware

It helps to place firmware among the three basic layers of a computer.

Layer	What it is	Example
Hardware	The physical parts of the computer	Processor, motherboard, keyboard
Firmware	Software fixed in non-volatile memory that controls the hardware	BIOS, UEFI
Software	Programs that the user runs or that run on the OS	MS Word, a web browser

Hardware is the body, software is the instruction set, and firmware is the low-level bridge that tells the hardware how to begin. Firmware is software by nature, but it is treated separately because it is stored on the device and is not installed or removed like an ordinary program.

46.4 Firmware on everyday devices

System firmware is not unique to computers. Many devices carry firmware that gives them their basic behaviour. A printer has firmware that controls printing; a Wi-Fi router has firmware that runs its networking; a smartphone has firmware plus an operating system; a modern television has firmware for its tuner and menu. Updating such a device usually means updating its firmware, again by flashing. This is why a firmware fix can repair a device without any change to the user’s applications.

46.5 System firmware: the BIOS

On a personal computer, the most familiar firmware is the **BIOS**, which stands for **Basic Input/Output System**. It is stored on a chip on the motherboard. When the computer is switched on, the BIOS runs first. It performs the power-on self-test, initialises the basic hardware, reads its settings, and then loads the operating system.

The single most important exam fact on this topic is that the firmware which initialises the hardware and loads the operating system is the BIOS. This is directly tested (PYQ-28). The BIOS is sometimes confused with a device driver or with the kernel; neither of these loads the operating system. A device driver is loaded *by* the operating system, and the kernel is part *of* the operating system.

Definition (BIOS). *The Basic Input/Output System: the system firmware on a PC motherboard that runs first at power-on, initialises the hardware, and loads the operating system.*

46.6 The power-on boot sequence

The machine follows a fixed chain each time it starts. Learning this chain makes the roles of firmware, the OS, and drivers easy to separate.

1. **Power on.** Electrical power reaches the motherboard.
2. **Firmware runs.** The BIOS or UEFI begins executing from ROM/flash.
3. **Hardware initialisation and POST.** The firmware tests and configures the basic hardware.
4. **Operating system loads.** The firmware finds the boot device, reads the bootloader, and hands control to the OS.
5. **Drivers load.** Once the OS is running, it loads the device drivers it needs.
6. **Applications run.** User programs start on top of the OS.

The chain can be written compactly as: power-on $\rightarrow$ firmware $\rightarrow$ initialisation $\rightarrow$ OS $\rightarrow$ driver $\rightarrow$ application. Any question that asks “which comes first?” is answered by walking this chain from left to right.

The step in which the firmware hands control to the operating system is worth a little more detail. The firmware locates the boot device according to its configured boot order, reads a small program called the **bootloader**, and runs it. The bootloader then loads the main part of the operating system into memory. Under the legacy BIOS, the bootloader is found through the MBR (Master Boot Record); under UEFI it is found through the GPT (GUID Partition Table) scheme and a dedicated EFI system partition.

Definition (Bootloader). *A small program that the firmware runs to load the operating system into memory and start it.*

46.7 Boot devices and boot order

The firmware does not decide by itself which device to start from. It follows the **boot order** stored in its settings: a list such as hard disk first, then optical drive, then USB. The firmware tries each device in turn until it finds one that can start the operating system. The boot order is the reason a machine can be made to start from a USB drive by moving that drive to the top of the list, and it is one of the first things a technician checks when a computer reports that no boot device is found.

Example 46.7.1. *A computer is switched on and shows, “No boot device found.” Walking the chain helps: the firmware runs, the POST passes, and the firmware then looks through the boot order for a startable device. If the disk has failed or its entry has moved down the list, no operating system is found and the firmware stops with this message. The failing layer is the firmware’s boot configuration or the boot disk, not the application software.*

46.8 POST: the power-on self-test

POST stands for **Power-On Self-Test**. It is one of the first things the BIOS does after power-on. During POST the firmware checks that the essential hardware — CPU, RAM, keyboard, disk controller, and video — is present and responding. If a component fails, the machine may beep in a code, show an error message, or refuse to continue.

POST is a step *inside* the firmware's work. It runs *before* the operating system, not after it. A frequent trap is to place the POST after the OS has started; the correct order is firmware (including POST) first, then the OS.

Definition (POST). *The Power-On Self-Test: the hardware check the BIOS runs at startup before loading the operating system.*

46.9 BIOS setup and CMOS

The BIOS keeps a small set of settings: the system date and time, the order in which the machine should look for a boot device, hardware options, and sometimes passwords. These settings are stored in a tiny battery-backed memory chip called **CMOS** (Complementary Metal-Oxide-Semiconductor). Because the chip is battery-backed, the clock and boot order are remembered between sessions, even when the main power is off.

The BIOS Setup utility (sometimes called the CMOS setup) is opened by pressing a key during startup. That key is commonly **Del** or **F2**, but the exact key varies by manufacturer. Inside setup, the user can change the boot order, enable or disable built-in devices, and set passwords.

46.10 UEFI: the modern firmware

UEFI stands for **Unified Extensible Firmware Interface**. It is the modern replacement for the legacy BIOS and performs the same basic job: it starts the hardware and loads the operating system. UEFI adds several improvements.

- It boots the machine **faster**.
- It supports very large disks through the **GPT** (GUID Partition Table) scheme, instead of the older MBR scheme with its practical limit of about 2 TB.
- It provides a **graphical** setup utility, often with mouse support.
- It offers **Secure Boot**, which helps prevent unauthorised software from loading at startup.

For exam questions, remember that “the firmware that boots the OS” may be either BIOS or UEFI depending on the age of the system. Both initialise the hardware and load the operating system; they are firmware, never drivers.

46.11 BIOS versus UEFI

The two system-firmware generations can be compared on a few clear points.

Point	BIOS	UEFI
Full form	Basic Input/Output System	Unified Extensible Firmware Interface
Age	Legacy (older)	Modern
Boot speed	Slower	Faster
Disk support	MBR (about 2 TB limit)	GPT (very large disks)
Interface	Mostly text-based	Graphical
Security	No Secure Boot	Secure Boot available

46.12 Device drivers

A **device driver** is software that lets the operating system communicate with a specific hardware device. Each device — a printer, a graphics card, a network card, a sound card — needs its own driver, because a driver translates the general commands of the operating system into the particular instructions that one device understands.

Drivers are loaded **by the operating system**, after the firmware has started the machine. They are matched to the exact device and the exact OS version. Because a driver is software, it is wrong to call it hardware, and because it is loaded by the OS, it is wrong to say that a driver initialises the hardware and loads the OS.

Definition (Device driver). *Software that enables the operating system to communicate with a particular hardware device by translating general OS commands into device-specific instructions.*

The three layers are easy to mix up, so keep their roles separate:

Layer	Stored where	Role
Firmware (BIOS/UEFI)	On the hardware (ROM/flash)	Starts the machine; initialises hardware; loads the OS
Device driver	With the OS, on disk	Talks to one specific device
Operating system	On disk	Manages all resources and provides the user interface

Example 46.12.1. *A question asks, “Which software initialises the hardware and loads the operating system?” A device driver is offered as an option. This is the classic near-neighbour trap. The correct answer is the **BIOS/UEFI** firmware. A device driver cannot do this job: it is itself loaded by the OS, and it controls one device rather than starting the whole machine.*

46.13 Firmware, driver, OS, and kernel

Four terms are often mixed up because they all describe “software close to the hardware”. Their roles are different, and the exam tests the differences.

Term	What it does	When it runs
Firmware (BIOS/UEFI)	Initialises hardware and loads the OS	First, at power-on
Kernel	The core of the OS; manages memory, processes, and devices	After the OS starts loading
Device driver	Communicates with one specific device	After the OS is running
Operating system	Manages all resources and provides the interface	After firmware, before applications

The kernel is part of the operating system, not separate firmware. A driver is loaded by the OS once the kernel is running. Only firmware runs before the OS. If an item asks which layer “starts the machine and loads the OS”, the answer must be firmware — BIOS or UEFI — and never a driver or the kernel.

46.14 Firmware updates and flashing

A **firmware update** replaces the firmware stored on a device with a newer version. The process is commonly called **flashing**, because the new firmware is written into flash memory. Updates are issued to fix bugs, add features, and patch security flaws.

Firmware updates are much rarer than ordinary software updates, and they carry more risk. Firmware is low-level, so a failed update can leave a device unusable — often described as “bricking” it. Interrupting the power during flashing, or loading the wrong firmware for the model, can damage the device permanently. For this reason the standard advice is to use only the manufacturer’s update for the exact model and to keep a stable power supply throughout the update.

Example 46.14.1. *A candidate reads, “A firmware update is called formatting.” This is wrong. Formatting prepares a storage volume for files; it does not rewrite firmware. The correct term for a firmware update is **flashing**.*

46.15 Exam retrieval and common confusions

The following distinctions prevent most errors on this topic.

1. Firmware that initialises hardware and loads the OS is **BIOS/UEFI**, not a device driver or the kernel (TRAP-24, PYQ-28).
2. A **device driver** is **software**; the OS is system software (TRAP-10).
3. Firmware is stored in **ROM/flash** and is non-volatile; RAM is volatile and empty at power-on.
4. **POST** runs **before** the operating system, as part of the firmware’s startup work.
5. A driver is loaded **by the operating system**, after the firmware has started the machine.
6. The startup order is power-on, firmware, initialisation, OS, driver, application.
7. A firmware update is called **flashing**; interrupting it can damage the device.

Example 46.15.1. *An item asks which layer runs first when a computer is switched on. Trace the chain: firmware runs before the OS, and the OS runs before any driver, which runs before any application. The answer is the firmware (BIOS/UEFI). If the stem instead asks what tests the hardware at startup, the answer is the POST; if it asks what lets the OS talk to one device, the answer is the device driver.*

Exercises

1. Define firmware and state where it is normally stored.
2. List the steps of the power-on boot sequence from power-on to application.
3. Expand BIOS, UEFI, and POST, and give one role of each.
4. Distinguish firmware, a device driver, and an operating system.
5. What is a firmware update called, and what is the main risk during one?

Solutions

1. Firmware is software stored in non-volatile memory that gives a device its basic, low-level startup and control instructions. It is normally stored in ROM or flash memory on the hardware itself.
2. Power on → firmware (BIOS/UEFI) runs → hardware initialisation and POST → the operating system loads → device drivers load → application programs run.
3. BIOS = Basic Input/Output System: system firmware that initialises hardware and loads the OS. UEFI = Unified Extensible Firmware Interface: the modern firmware with fast boot, GPT disk support, and Secure Boot. POST = Power-On Self-Test: the hardware check the BIOS runs at startup.
4. Firmware is stored on the hardware in ROM/flash and starts the machine and loads the OS. A device driver is software loaded by the OS that talks to one specific device. The operating system is the large system-software layer that manages all resources and provides the user interface.
5. A firmware update is called flashing. The main risk is that interrupting the power during flashing, or using the wrong firmware, can leave the device unusable.